ILLINOIS STATE GEOLOGICAL SURVEY
PRAIRIE RESEARCH INSTITUTE

Final Report of TDLAS CO_2 Sensor Field Testing at the Illinois Basin – Decatur Project

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EXECUTIVE SUMMARY

This report provides a summary of the investigation of prototype infrared diode laser-based sensors intended to continuously monitor carbon dioxide (CO_2). Their principal purpose is to discern anomalous temporal or spatial changes resulting from local CO_2 emissions that may indicate leakage from carbon capture and storage (CCS) reservoir sites or related pipeline and wellhead infrastructure. The initial applications are intended to be for personnel safety and pipeline leak inspection. The sensors also have potential beneficial uses for environmental monitoring. The prototype equipment was designed by Physical Sciences, Inc., and was field tested in Champaign, Illinois and Decatur, Illinois as part of the Illinois Basin – Decatur Project (IBDP) being led by the Illinois State Geological Survey (ISGS). Three sensor configurations based on infrared tunable diode laser absorption spectroscopy (TDLAS) were investigated: (i) an Open-Path Sensor (OPS) for CO_2 which measures path-integrated carbon dioxide concentrations over a fixed path to monitor along pipelines and at wellheads, (ii) a Remote Carbon-Dioxide Leak Detector (RCLD) which is a compact and mobile diagnostic tool intended for pipeline surveying, and (iii) an In-ground Sensor (IGS) intended for monitoring carbon dioxide levels in shallow wells or subsurface agricultural drainage systems. Two of these sensors, the OPS and RCLD were deployed and tested at the Illinois Basin - Decatur Project (IBDP) site in Decatur, Illinois. Efforts towards constructing and deploying the IGS were made, including delivery of several components, layout of power and data collection support equipment, and plans for field deployment. However, a specialized infrared laser component was not functional and a suitable replacement could not be obtained before the project ended. Therefore, the construction of the IGS configuration could not be completed. This report includes a summary of project goals, TDLAS theory, components of each CO_2 sensor system, and key results obtained in laboratory and field testing. The challenges associated with each diagnostic are discussed, and lessons learned are summarized. The collaborative investigations of Physical Sciences Inc. and the Illinois State Geological Survey for this program have illustrated the uses of TDLAS-based sensors. More specifically, two types of CO_2 sensor and monitor configurations were tested successfully at the IBDP site. Both demonstrated detection and identification of abnormal CO_2 releases from wellhead pipelines. With refinements, similar systems could be used at other CCS sites as part of an integrated leak detection and environmental monitoring strategy.

I. PROJECT INTRODUCTION

The report is a summary of a U.S. Department of Energy (DoE) funded Small Business Technology Transfer (STTR) Phase II project extension in which the Illinois State Geological Survey (ISGS) served as a subcontractor to Physical Sciences Inc. (PSI). The performance period of the project extension was June 16, 2012 through November 14, 2013. The goal of the ISGS investigation was to demonstrate prototype infrared diode laser-based sensors at an active CCS project in Decatur, Illinois, USA. These types of sensors have promising potential to be used for CO_2 leakage detection as part of environmental monitoring programs at carbon sequestration sites.

A. Overview of the Illinois Basin Decatur Project

The Midwest Geological Sequestration Consortium (MGSC, see www.sequestration.org [1]) is conducting a demonstration-scale CCS project in Decatur, Illinois, USA to test the ability of a deep saline formation to store one million tonnes of carbon dioxide (CO_2) from an ethanol production facility. In November 2011, CO_2 injection began at a rate of 1,000 tonnes per day and will continue for three years into the Mount Simon Sandstone, a deep saline reservoir, at a depth of approximately 2,100 meters (7,000 feet).

An extensive Monitoring, Verification, and Accounting (MVA) program has been undertaken for the Illinois Basin – Decatur Project (IBDP) and is focused on the 0.65 km^2 (0.25 mi^2) project site. Near-surface and subsurface monitoring are integral efforts to reach MVA and project goals of 1) establishing pre-injection conditions to evaluate potential impacts from CO_2 injection, 2) demonstrating that project activities are protective of human health and the environment, and 3) quantifying and tracking CO_2 stored in the Mount Simon Sandstone reservoir. The IBDP MVA Program is a combined effort of the ISGS, Archer Daniels Midland (ADM) Company, Schlumberger Carbon Services (SCS), Schlumberger Water Services, PSI, Lawrence Berkley National Laboratory, University of Illinois, TRE-Canada and the Carbon Capture Project, and the Illinois Department of Transportation. More than 20 different monitoring methodologies are being used at IBDP to field test the efficacy of those methods to monitor large-scale CCS projects.

Near-surface monitoring efforts include near-infrared color aerial imagery acquisition, surface deformation monitoring, net CO_2 flux monitoring, soil CO_2 flux monitoring, soil gas sampling, high-resolution electrical earth resistivity surveys, and shallow groundwater sampling. Subsurface monitoring efforts include 2D seismic surveying; 3D seismic and vertical seismic surveying; passive seismic monitoring; injection zone temperature, pressure, and fluid monitoring; above caprock temperature, pressure, and fluid monitoring; and open and cased hole logging. Environmental monitoring was initiated in 2009 and is currently scheduled to conclude in 2017 after the three-year injection and three-year post-injection periods have concluded. On-going injection operations are being conducted under a Class I – non-hazardous underground injection control (UIC) permit issued by the Illinois Environmental Protection Agency. A Class VI UIC permit application was submitted to the U.S. Environmental Protection Agency in December 2011 for the IBDP and is currently under review.

B. Summary of ISGS Efforts

In chronological order, the planned ISGS efforts to support PSI on this DoE STTR project included:

1. Identification and characterization of suitable locations for deployment of prototype sensors including an additional agricultural subsurface drainage investigation if needed.

2. Coordination of site selection, finalization, and approval with PSI and ADM.
3. Hosting an IBDP field visit of project participants to select deployment locations.
4. Acquisition of local materials and services for field installations of monitoring equipment and site characterization.
5. Preparation and construction of selected field sites with PSI staff input for deployment of the open-path and in-ground sensors.
6. Sensor deployment in coordination with other IBDP project activities and MGSC partners (e.g., ADM).
7. If needed, minor maintenance of open-path and in-ground sensors after deployment.
8. Periodic communication with PSI including site/sensor conditions and data transmissions to PSI.
9. Supporting of project personnel that will collect data from the installed sensors, work and train with PSI to process/analyze field data from the deployed sensors, learn sensor construction details from PSI to facilitate field deployments, and assemble additional deployable sensors subject to availability.
10. Conducting a field test of the hand-held, portable sensor subject to prototype availability.

C. TDLAS Prototype Device Configurations

For this project, PSI focused on building and testing three prototype sensor configurations for leak detection using tunable diode laser absorption spectroscopy (TDLAS). The sensor configurations included:

1. An "open-path" sensor (OPS) that continuously monitors CO_2 levels along a line of sight approximately 100 m long, intended as an alarm-type system for personnel safety that would be used along pipelines and at wellheads;
2. An "in-ground" CO_2 point sensor (IGS) intended to provide long-term continuous measurement of CO_2 concentrations and changes over time from a shallow subsurface installation (e.g., agricultural drain, groundwater monitoring well); and
3. A remote carbon dioxide leak detector (RCLD) that is a hand-held, portable device intended for CO_2 pipeline leak surveying and detection.

The sensors were intended to 1) provide CO_2 monitoring data from field equipment with limited maintenance needs and 2) be operated by end users with minimal need for advanced operational training. The IBDP site was an excellent location for field testing, because it had an extensive environmental monitoring program already in place that would complement sensor testing.

II. TDLAS Concept and Theory

The three sensor systems discussed here all apply TDLAS using infrared semiconductor lasers and photodetectors to measure the infrared absorption of CO_2. The method takes advantage of the fact that CO_2 molecules absorb infrared (IR) radiation on well-known, narrow-wavelength bands centered on specific wavelengths in the infrared electromagnetic spectrum. The PSI sensors operate in the near-IR at wavelengths in the 1.5 µm and 2.0 µm spectral regions.

Figure 1 shows the basic layout of a TDLAS sensor. The simple layout consists of 1) a tunable infrared laser which is tuned in its center wavelength and intensity by modulation of the current; 2) an optical path which traverses a region bearing the IR absorbing gas of interest (in this case CO_2); and 3) a photodetector. Figure 2 illustrates the basic signal relationships used in determining absorption along the optical path of the laser. The laser current is modulated at a frequency (f_m) resulting in a modulation of

intensity at the detector, creating a component of the detector signal at frequency f_m. This signal component is called F_1. As the laser *current* modulates, the laser *wavelength* also modulates over the line shape of a CO_2 absorption feature. During a single current modulation sweep, the laser wavelength passes through the shape of the absorption feature twice, resulting in a component of the detector signal amplitude modulation at $2*f_m$. This signal component is called F_2. The OPS signal processor demodulates the periodic signal to measure the RMS amplitudes of F_1 and F_2, and these components are used to calculate the CO_2 concentration. F_1 is a measure of the laser power received at the detector; it is generally free of influences from other sources of optical power, such as sunlight, that impinge of the detector. F_2 is proportional to the product of received laser power and path-integrated CO_2 concentration. Thus, the ratio F_2/F_1 provides a measurement that is directly proportional to CO_2 and is independent of received laser power. This feature enables the sensors to correctly track CO_2 despite changes in laser power transmitted across the optical path. As seen in some of the data illustrations below, such changes occur due to variability in reflectance of illuminated backscatter surfaces, precipitation in the optical path, thermally-induced movements of optical components affecting alignment, and the like. The proportionality constant that converts the measured F_2/F_1 to CO_2 concentration is generally determined by calibrating the sensor with a known amount of CO_2 in the laser path.

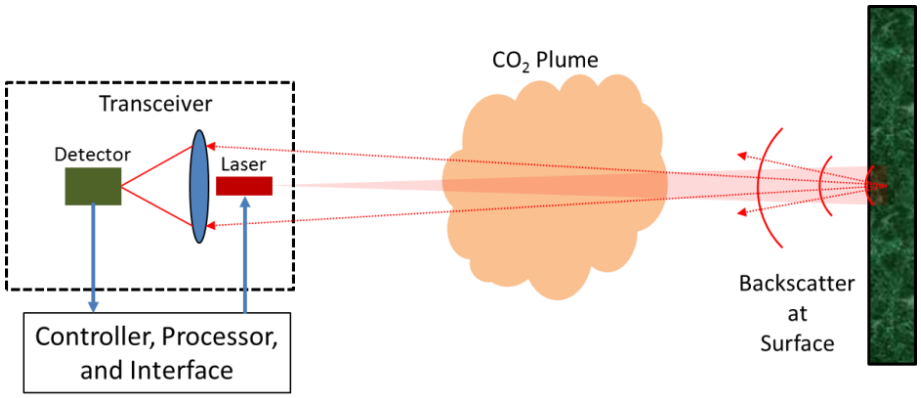


Figure 1. Illustration of the TDLAS sensor concept.

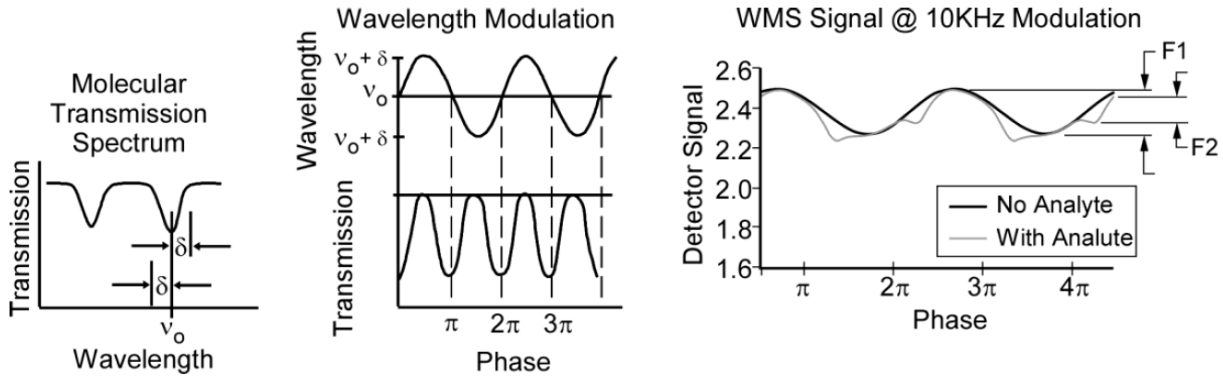


Figure 2. Wavelength Modulation Spectroscopy (WMS): Laser wavenumber (ν) modulates across a spectral absorption line with modulation depth (δ) creating a modulated transmitted signal at the detector. Taken from: M. Frish, et al. "Precision and Accuracy of Miniature Tunable Diode Laser Absorption Spectrometers." Physical Sciences Inc., Report SR-1446 [2].

III. OPEN-PATH SENSOR

This section is a summary of the deployment experiences of the OPS at the IBDP site in Decatur, IL. The OPS uses near-infrared TDLAS to measure CO_2 absorption over a 100 m path-length, situated immediately north of the injection well/pipeline, with the path running east (laser transceiver) to west (reflector). Thus, the laser passes over the 100-m path twice, being launched from the transceiver toward the reflector, and then returning along the same path to the transceiver collection optics. The OPS transceiver continuously transmits CO_2 absorption data via radio to the data-logging workstation setup in the trailer to the south of the injection well.

The OPS was initially deployed at the IBDP site on January 7, 2013 and is planned to be retrieved in December 2013. An aerial view of the deployment location is shown in Figure 3. Because only relative CO_2 values need be measured to perform the intended leak detection function, the sensor was not fully calibrated upon deployment; the installed calibration constant was an approximately value based on prior testing at the PSI facility. As discussed below, near the completion of the test period data were acquired with an alternate CO_2 sensor, thus enabling OPS calibration to yield absolute CO_2 values.

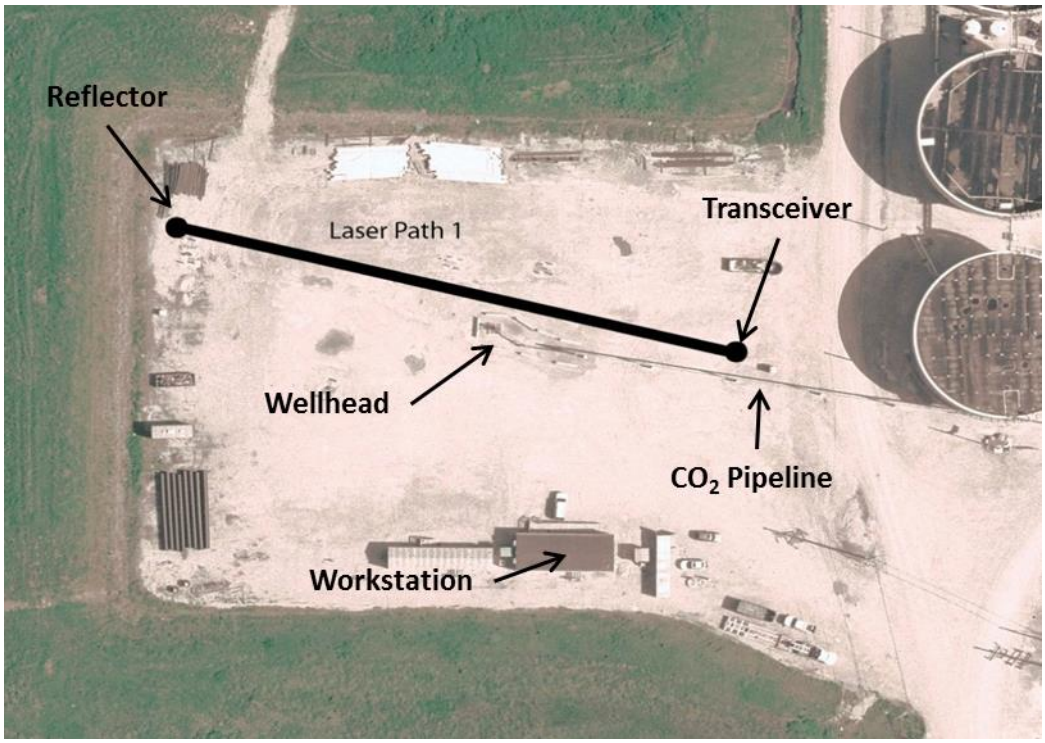


Figure 3. Location of PSI TDLAS beam line for field testing of prototype CO_2 Open-Path Sensor (OPS)

A. OPS Operation and Components

The PSI OPS takes advantage of the near-infrared absorption spectra of the CO_2 molecule in order to determine CO_2 levels along its optical path. The device has four primary components:

- A fiber coupled tunable infrared diode laser,
- Primary optics which launch the fiber-coupled laser beam along the absorption path,

- A passive directionally-reflective target (fabricated from a commercial Stop sign partially covered with reflective tape) at the opposite end of the absorption path which returns laser radiation back to the transceiver, and
- Secondary optics which collect the returned laser radiation on to an infrared sensor.

Additional components to the system include (i) a lithium ion battery which is periodically recharged by a solar panel and supplies power to the transceiver components, (ii) a wavelength modulation spectroscopy circuit board which controls the tunable laser, and analyzes the sensor data, (iii) a wireless radio modem which transmits the board data to a workstation, and (iv) a workstation which collects the transmitted circuit board data, performs calculations, and logs data for later use and inspection.

B. OPS Activity Timeline

Table 1 provides a timeline of deployment and testing activities for the OPS in the laboratories in Champaign (at the ISGS) and in Decatur (at the IBDP site).

Table 1. Summary timeline of OPS deployment and testing

Date(s)	Tasks
11/9/2012	Meeting at IBDP injection site between Curt Blakley (ISGS), Scott McDonald (ADM), and Jim Kirksey (SCS) to review instrument location and laser path orientation.
11/25/2012 to 11/30/2012	Trip to PSI facility in Andover, Massachusetts for laser training and equipment transport. Relocated equipment to ISGS in Champaign, IL
12/12/2012	Installation of the OPS footing, solar panel post, and metal bollards using the ISGS Direct Push Power Probe
1/7/2013	Initial transceiver and reflector deployment
1/9/2013	Initial alignment and setup of radio data logging (OPS subsequently retrieved from the site due to performance issues)
1/17/2013	Laboratory troubleshooting of equipment in Champaign. Improved weather proofing. Redeployed at IBDP in the afternoon.
1/22/2013	Checked alignment, radio/software operation
1/25/2013	Setup desktop workstation at IBDP for remote data logging
1/30/2013	Issues with insufficient battery power encountered (after work at the ISGS, and insufficient solar charging)
2/1/2013	Reinstalled battery (charged w/DC), measured signals vs. distance (aluminum tape reflector).
2/6/2013	Performed “bump” test w/ CO_2 bag, measured signals vs. distance (stop sign w/ reflective tape).
2/8/2013	Removed cardboard baffle (slight increase in F1), aligned and focused using F1 signal
2/11/2013	Used CO_2 bag to map positions of IR beam on stop-sign reflector, compared to red sighting laser
2/15/2013	Performed “bump” tests at 10, 20, 50, 100 m.
2/18/2013	Removed reflective tape from bottom, top portions of sign, leading to significant increase in F1 and F2 signals.
3/2013 to 5/2013	Performed periodic routine maintenance, data file collection, and instrument alignment. Assessment of battery and power issues conducted.
5/20/2013	Collected battery voltage and current draw, panel power data (sunny conditions)
5/31/2013	Collected battery voltage and current draw, panel power data (cloud cover varies)
6/18/2013	Adjusted OPS radio output wattage from 1W to 100 mW to reduce power consumption.
6/18/2013 to 11/15/2013	Performed periodic routine maintenance, data file collection, and instrument alignment. Assessment of instrument field capability to withstand moisture and temperature variations.
10/1/2013 to 11/15/2013	Collected periodic atmospheric CO_2 concentrations with LI-COR 8100A (infrared gas analyzer) to compare with synchronous OPS measurements
10/18/2013	Replace 30 watt (26.2” x 16.2”) solar panel with larger, 65 watt (37.1” x 19.7”) solar panel

C. Challenges of Deployment

During OPS testing, a few challenges were encountered and are summarized below.

Power Supply Issues (Budgeting)

Despite PSI reporting nearly one-year of solar-powered operation during testing of the OPS in Andover MA, when installed at the IBDP site the unit initially experienced intervals of insufficient charge causing loss of data. Given the known transceiver system (laser, transceiver board, radio, etc.) design specifications, the sensor power requirements are well-defined, but variable weather conditions made it difficult to specifically predict the power available for the solar panel-based recharging system. Shadows from adjacent infrastructure also reduced sunlight availability for system charging.

During spring 2013, the 19 amp-hour (AH) battery (Southwest Electronic Energy Group #7719069 Four Series {SWE www.swe.com}) used in the OPS was not well sustained by the 30-W solar panel installed initially. Using recommended equipment from PSI, the voltage and current data of the power supply system were analyzed. LASCAR USB voltage data loggers (EL-USB-3) were used to monitor the voltage at the panel input and the battery output terminals. A Sperry DM-6450 digital multimeter was used as an ammeter to measure the current between the solar panel and the OPS battery, and the current from the battery to the laser/sensor board and radio (at varied data logging rates). Table 2 shows representative data of system voltages and currents for various cloud cover conditions, along with comparison of the panel power to the power used by the OPS (assuming a 12-hour charging period). The data indicated that the panel power was not sufficient to sustain OPS through extended periods of heavy cloud cover. To resolve this condition, the 30W solar panel was replaced by a 65-W panel on 10/18/13.

Table 2. OPS voltage and current measurements for varied cloud-cover.

Date	Conditions	V_{BAT} [V]	V_{PAN} [V]	I_{PAN} [A]	P_{PAN} [W]	$P_{PAN} / 2P_{OPS}$
5/20/13	Sunny	15.08	15.94	1.15	18.3	1.84
5/31/13	Light Clouds	14.45	15.20	0.68	10.3	1.03
5/31/13	Moderate Clouds	14.45	15.20	0.53	8.06	0.81
5/31/13	Heavy Clouds	14.45	15.20	0.40	6.08	0.61

Thermo-mechanical Alignment Drift

TDLAS applications require well-designed optics. Factors that must be considered are path length, the beam divergence and far-field shape, the collection optics field-of-view, reflector sizing, and the rigidity of the mechanical components. Various structural components flex due to heating, which leads to changes in alignments of the optical components. During installation, components (specifically aiming of the laser at the reflective target) are aligned to optimize the received laser power. Any subsequent changes in alignment cause received laser power to diminish from the optimum. As described previously, in general the CO_2 measurement is unaffected by these changes. However, there is a minimum amount of laser power that must be received to preclude noise and errors in the CO_2 measurement. The sensor automatically recognizes when the received power is too small for proper measurement. In its current configuration, when the F1 signal is too small an error code is written in the data file, but CO_2 data is still reported, albeit possibly erroneously.

The current version of the OPS prototype uses a modified commercial telescope with the fiber collimation optics and a launch reflector mounted at the front of the telescope and a sensor mounted in place of the

telescope objective. While this configuration served well for demonstration purposes, a design similar to that applied in the RCLD would be more robust and is recommended. With this modification the OPS could serve as a more robust alarm sensor, and potentially also as a calibrated diagnostic for environmental monitoring applications.

Environmental Factors and Scheduling

A few environmental factors were observed during the OPS deployment and should be considered in future OPS deployments including:

- Interruption of data due to vehicles, people, animals, etc. blocking the IR beam path. If possible, signs should be posted to identify the path and limit beam interruption. Elevating the laser path above traffic may help limit these interruptions.
- Structures on-site should be taken into consideration when positioning the solar panels such that good sun exposure is guaranteed throughout the year, and a schedule for repositioning the panel as necessary should be made. At the IBDP deployment location, an overhead pipeline periodically cast a shadow on the OPS solar panel reducing available sunlight for proper system charging. In appropriate locales, solar power may be supplemented or replaced by wind power.
- Firmness of the reflector/target installation: there were some issues with wind knocking down the metal-sign reflector used for the OPS. This could be remedied using a more permanent (concrete) or more stable (larger base) installation.
- Radio data-logging: The radio power level and antenna system should be set up such that clear, uninterrupted communications are established between the OPS transceiver and the data logging workstation.
- Clear communications were needed with facilities managers and pipeline operators to identify variations in absorption that were due to normal operation (as opposed to actual unintended releases). At the IBDP site, no unintended releases occurred, but several routine releases (i.e., CO_2 pipeline blowdowns prior to startup of injection operations) occurred and were used as signals of opportunity to test the OPS responses. In addition, frequent communications with facilities personnel were important to adequately understand and document interruptions in OPS signals that were related to physical blockages (i.e., parked vehicles) that interrupted the sensor beam.

D. Data Summary

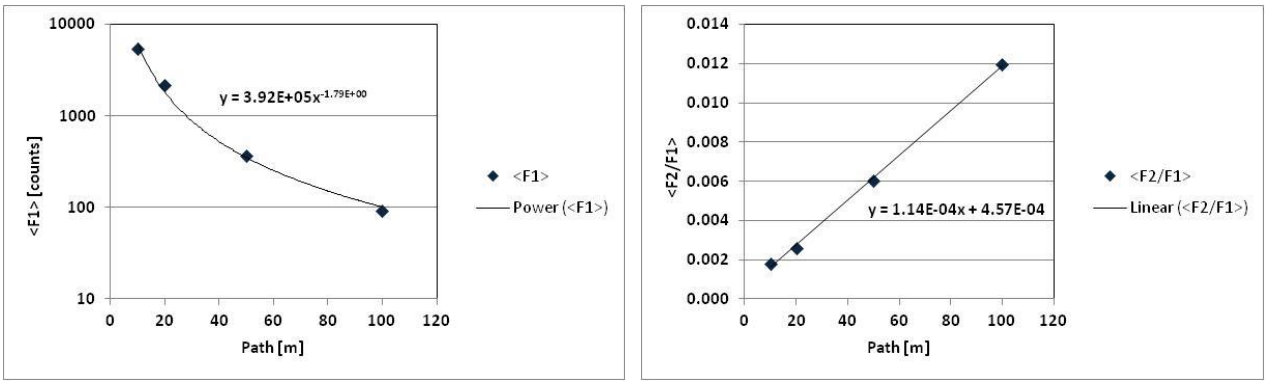
A summary of concentration measurements taken throughout the evaluation deployment at IBDP is given in Appendix A. In this section, selected results are summarized that illustrate some typical behaviors that were observed.

Initial Testing

Some initial tests were performed after deployment to verify that the OPS would respond to CO_2 in its path using an infrared transparent bag filled with CO_2 at the reflector surface and multiple absorption path distances. Figure 4a shows measurement of F1 as a function of path length (taken 2/6/13). As expected, the signal diminishes with distance (~path^{-2}). Figure 4b shows a linear increase in the ratio F2/F1 with path length. As described above, this ratio is proportional to path-integrated CO_2 concentration; its linear increase with distance indicates proper response to sensing ambient CO_2. The data in Figure 5 shows the linear response of <F2/F1> with distance

and the increased response of the OPS when the CO_2 absorber bag is placed in the beam path (i.e., a "bump test").

In these initial tests, the F_1 signals with the reflector at 100 m were lower than desired (< 100), and were below the desired threshold used in the concentration calculation. During the alignment procedure, it was noted that higher F_1 and F_2 signals were achieved when the laser was aligned off the Stop-sign center, such that it was partially overlapping a portion of the Stop sign without reflective tape. Based on this, it was concluded that the Stop-sign surface itself was acting as a better IR reflector than the reflective tape. On 2/18/13, a considerable portion of the reflective tape was removed from the sign (leaving a portion covering the word "stop"); this resulted in a significant increase in F_1 and F_2 signals (by roughly a factor of 5).



(a) <F1> as a function of path length. **(b)** <F2/F1> as a function of path length.

Figure 4. OPS response to different path lengths up to 100 m (data from 2/6/2013).

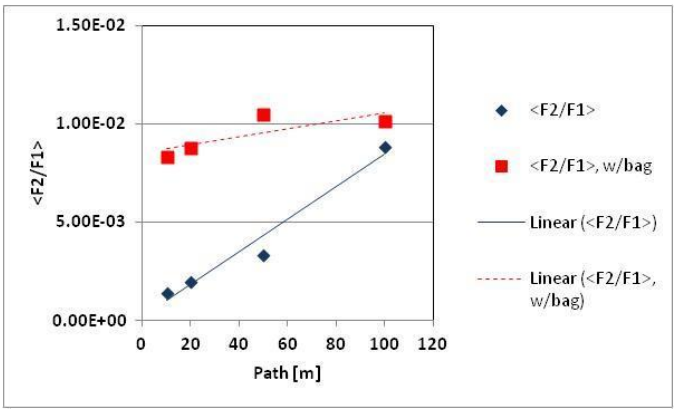


Figure 5. <F2/F1> as a function of path length without and with CO_2 bag at center of reflector (data from 2/15/2013).

Weather Effects

The OPS was operated during a variety of weather conditions (rain, snow, high winds, etc.). Figure 6 shows an example of the influence of fog and mist on OPS signals. Despite a significantly reduced return signal due to increased broadband attenuation, the measurement of CO_2 level was maintained.

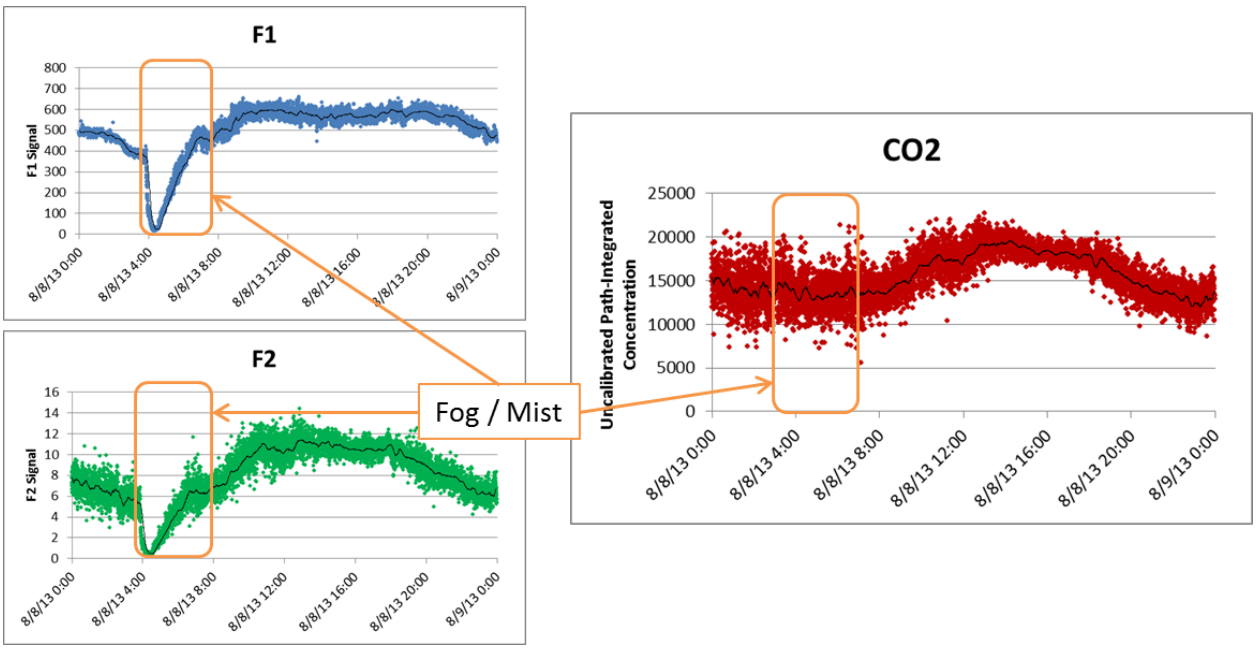


Figure 6. OPS readings during fog and mist (8/8/2013).

OPS Performance during Routine Pipeline Blowdown

Maintenance operations were performed on the IBDP injection well on 9/6/2013, and it was possible to test the OPS responses (Figure 7) with exposure to CO_2 concentrations greater than atmospheric background. A routine pipeline blowdown procedure elevated CO_2 concentrations for approximately one hour in the vicinity of the OPS installation before CO_2 injection operations resumed. Similar results are overviewed in Appendix C.

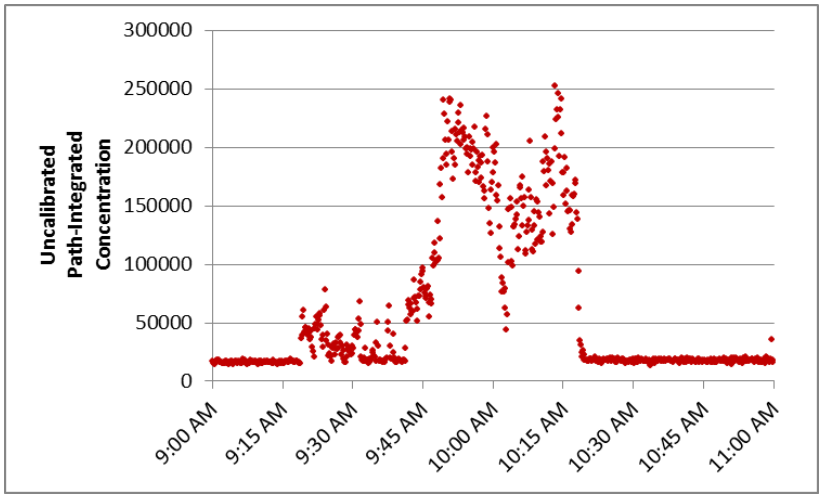


Figure 7. OPS CO_2 measurements during routine IBDP pipeline blowdown (9/6/2013).

Comparisons of OPS to LI-COR Measurements

During recent months, the OPS was compared to atmospheric CO_2 measurements with LI-COR devices operated at the IBDP site - the LI-COR LI-8150 multiplexer, and the LI-COR LI-8100A Soil CO_2 Flux System.

The LI-8150 multiplexer is a soil flux monitoring system used at the IBDP site to characterize temporal variations of soil CO_2 fluxes at multiple locations. It is installed in a fixed location in a field approximately 350 m northwest of the OPS absorption path. Four soil flux rings (two natural and two treated with herbicide) and four atmospheric sampling ports collect soil flux and atmospheric CO_2 measurements on a thirty minute sampling cycle. Atmospheric ports 5, 6, 7, and 8 are set at 0.3, 1.8, 5.5, and 8 feet above the ground, respectively. Figure 8 shows the OPS data for August-October 2013 compared to the LI-COR LI-8150 measurements for atmospheric port 7. The data shown for the OPS is the integrated concentration in ppm-m divided by the path length (100 m).

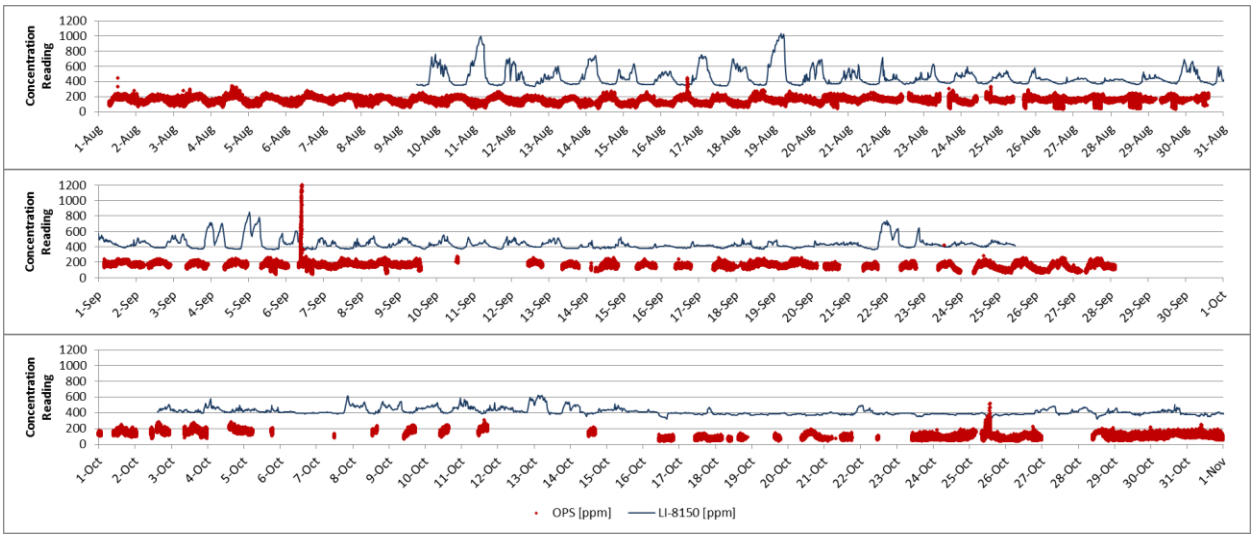


Figure 8. Comparison of OPS readings with LI-COR LI-8150 measurements during August-October 2013.

The variations in the multiplexer data are irregular, with the tendency of strong peaks in the night time and valleys during the daytime hours. The OPS measurements show an opposing behavior, typically peaking during daytime hours and reaching minimums at night. The differences observed may be related to the differences in the surficial conditions near the monitoring locations. At the OPS location, there is no vegetation within 50 m of the instrument. At the multiplexer location, the instrument is surrounded by a field of vegetation.

During eight visits to the IBDP site in October 2013, ISGS staff made measurements comparing the OPS to concentration measurements from a LI-COR LI-8100A at three positions along the OPS absorption path. Results of this comparison are in Figure 9.

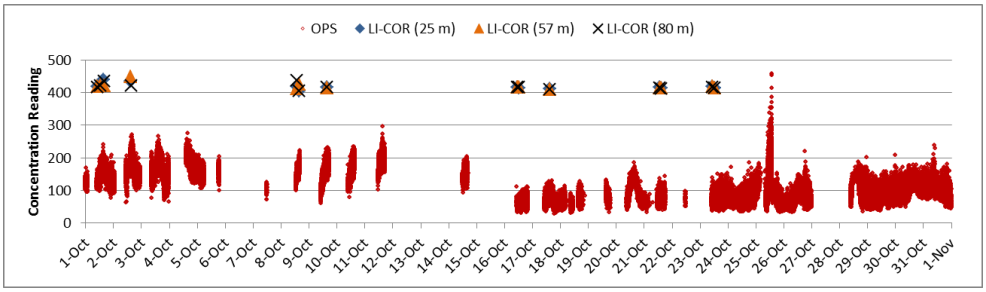


Figure 9. Comparison of OPS readings with LI-COR LI-8100A measurements at various distances from the transceiver (westward) along the absorption path.

In comparison to both LI-COR diagnostics, the concentrations measured by the OPS were significantly lower, indicating that OPS calibration is needed so the OPS signal can be tuned to provide more accurate CO_2 measurements. These comparisons may be used for OPS calibration; they indicate that dividing OPS ppm-m measurements by a calibration constant of 40 will yield path-averaged CO_2 concentrations (ppm) consistent with the LI-COR values.

E. OPS Performance Observations

Given below is a summary of significant observations and lessons learned.

- As expected, the OPS concentration measurement (ppm-m) scaled linearly with distance while F1 decreased as d^{-2}.
- The OPS responded promptly to changes in CO_2 during routine pipeline blowdown, illustrating its utility as a permanent monitor to detect and report unintended leakage.
- The OPS was not calibrated to report absolute CO_2 concentrations. CO_2 values calculated by the real-time software were typically a factor of 2 smaller than comparison measurements acquired with LI-COR instruments.
- Installation of the wireless solar-powered transceiver was straightforward and simple.
- The procedure for aiming the OPS laser beam at the reflective target, aided with a 50 milliwatt 650 nm fiber-coupled "fault-locator" attached to the launch fiber in place of the IR laser, was straightforward over the 100-m path.
- The 30-W solar panel was insufficient for sustaining power to the OPS in poor weather conditions and was replaced with a 65-W panel (10/18/13).
- The 19.3 Ah Li-ion battery selected for the OPS performed well throughout the evaluation at IBDP, despite numerous charging and discharging events due to lack of charging current in poor weather conditions.
- Low return signal, believed due to thermo-mechanically induced alignment degradation of optical components, occasionally resulted in loss of data. Launch and collection optics designed for reduced thermal sensitivity would enhance system autonomy. (PSI notes that this modification has been made to an OPS methane sensor being demonstrated elsewhere, and indeed demonstrates significantly less alignment drift over ~200m laser pathlengths.[1])

IV. IN-GROUND SENSOR

A. In-Ground Sensor Components

Sensor Module

The sensor module parts for the in-ground sensor (IGS) system were shipped to the ISGS the week of April 29, 2013.

Figure 10 shows an assembly drawing of the in-ground sensor head. The IGS utilizes the same control and signal processing electronic board as the OPS and RCLD (described below). However, unlike those sensors which utilize mature distributed feedback (DFB) fiber-coupled diode lasers in 14-pin butterfly

[1] M.B. Frish and B.A. Cummings, "Laser Sensors for Gas Pipeline Explosion Protection", Draft Final Report prepared for the California Energy Commission, PSI 1802-TR-2926, November 2013.

packages installed on the electronics board, the IGS utilizes relatively new 2.0 µm VCSEL (**v**ertical-**c**avity **s**urface-**e**mitting **l**asers) in TO-39 packages installed within the sensor head. PSI had not previously utilized these VCSEL products in their laser sensors.

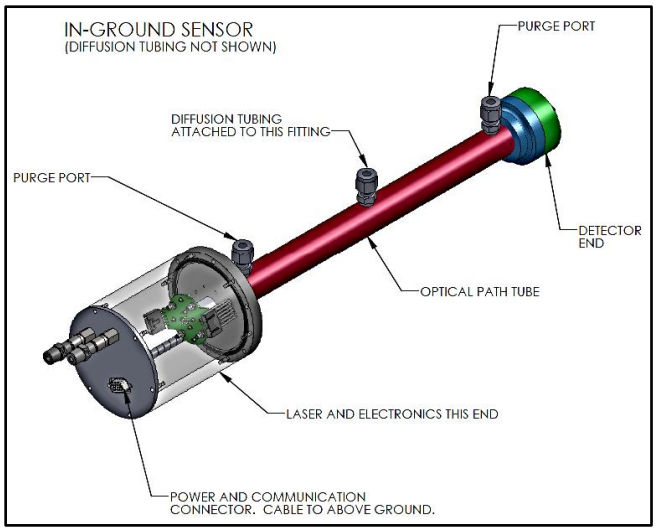


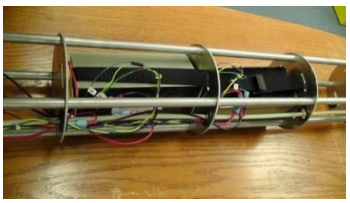
Figure 10. Assembly drawing of the In-Ground CO_2 Sensor (IGS) [3].

Support Equipment for Deployment

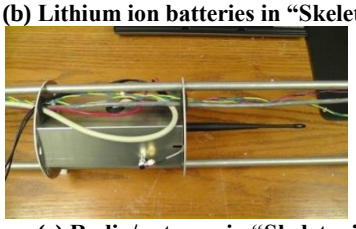
The IGS is intended to operate in much the same way as the OPS, requiring a solar-panel charged battery to power the sensor and a radio system to communicate between the sensor and a control computer near the deployment site. The configuration chosen for providing power was a modified version of the Heath Consultants "Dig In" system shown in Figure 11, which stores the lithium ion batteries and radio inside a sealed PVC tower.



(a) Power/communications tower w/ solar panel



(b) Lithium ion batteries in "Skeleton"

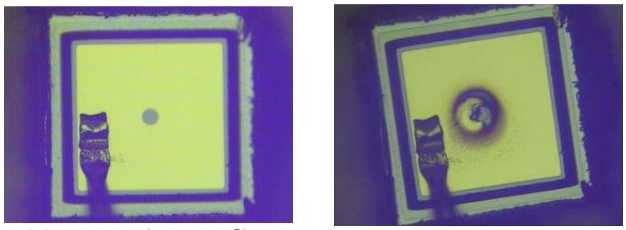


(c) Radio/antenna in "Skeleton"

Figure 11. The Heath Consultants power supply and communications tower suggested for use with the in-ground sensor [4].

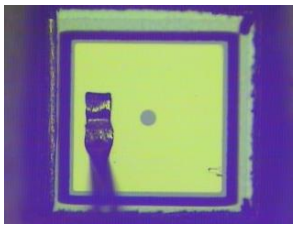
B. Analysis of Transistor Outline-style Laser Failure Issues

All of the components required to assemble the sensor head with the exception of the infrared diode laser component were delivered to the ISGS. Subsequently, one of two VCSELs used in the design was delivered to the ISGS by PSI. However, the VCSEL was nonfunctional when delivered to the ISGS, having failed in the PSI laboratory just prior to sensor shipment after several months of operation. The second VCSEL had previously failed in similar fashion. PSI contacted the VCSEL supplier concerning the failure issue in mid-May 2013 and returned the failed devices on July 11, 2013. The supplier analyzed the devices and concluded that the lasers had failed due to electrostatic discharge, evidenced by the visual and electrical characteristics shown in Figure 12. In the case of VCSEL #1, there was obvious damage to the laser aperture. In the case of VCSEL #2 the damage was indicated by the ohmic behavior (red curve) of the U-I curve

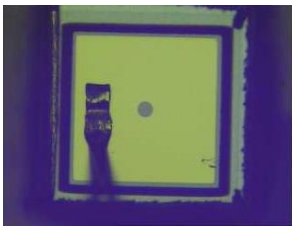


VCSEL #1 Prior to Delivery to PSI

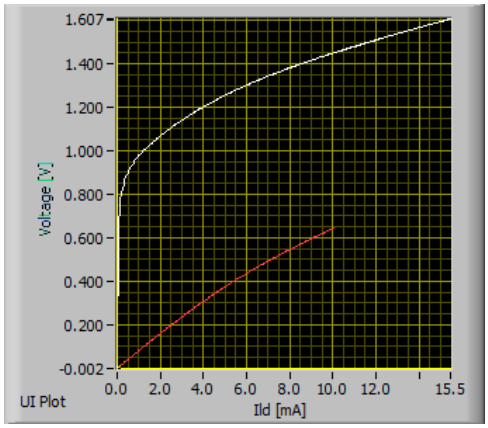
VCSEL #1 RMA Report



VCSEL #2 Prior to Delivery to PSI



VCSEL #2 RMA Report



U-I Curve for VCSEL #2. The white curve is prior to delivery, the red curve was measured for the RMA report.

Figure 12. Analysis of failed VCSELs (Vertilas) [5].

During August 2013, the ISGS contacted several companies in search of an alternate option for the 2.0 µm VCSEL. None were found.

The failure analysis indicates a potential incompatibility among this particular VCSEL product, the sensor configuration, and PSI laser control electronics. Circuitry modifications to further suppress susceptibility to ESD are likely to resolve the problem. Without such modifications, confidence in a successful in-ground sensor deployment that will yield long-term data is low. Unfortunately, replacement VCSELs require 3-6 months for delivery, and must be followed by several months of laboratory testing to verify that the sensor modifications were successful. Therefore, final construction of the IGS could not be completed during the project period and the instrument was not deployed at the IBDP site.

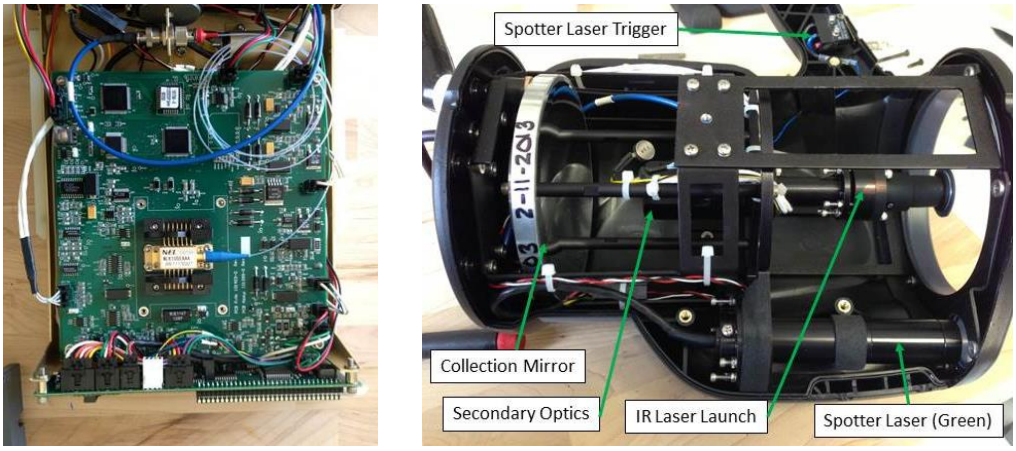
V. Remote Carbon-Dioxide Leak Detector

This section summarizes the modification of a Heath Consultants Remote Methane Leak Detector (RMLD) to measure carbon dioxide absorption, and the laboratory and field testing of the modified device.

A. Modification of the RMLD to Measure CO_2 Concentration

During March 19 to 22, 2013, ISGS staff (J. Zimmerman) traveled to PSI in Andover, MA to work with PSI staff (M. Frish, M. Laderer, and R. Wainner) on modification of a Heath RMLD ([6]) for detection of CO_2, to develop a fabrication and testing plan for the prototype IGS, and to receive training on operation and maintenance of both devices.

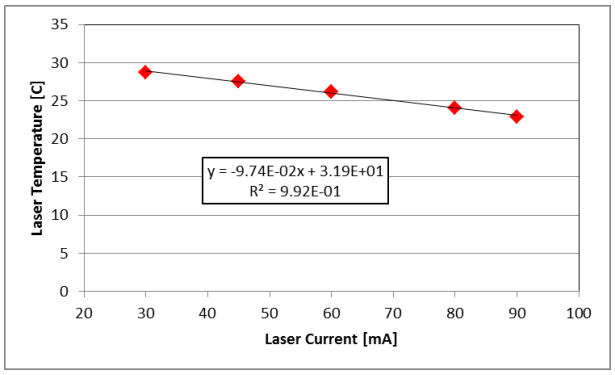
Modification of the RMLD for CO_2 detection involved: a) substituting the RMLD's methane-specific 10mW 1.6µm DFB laser with a CO_2-specific 3mW 2.0µm DFB laser in a drop-in replacement fiber-coupled 14-pin butterfly package; and b) replacing the RMLD's InGaAs detector and pre-amp with an extended InGaAs detector and matched pre-amp sensitive to 2.0 µm light. This combination of laser and detector installed in the RMLD transceiver was expected to provide a useable standoff range of about 10m when aimed at a diffuse (i.e., non-reflective) scattering surface, as compared to the standard RMLD which typically provides standoff detection range up to 30m.



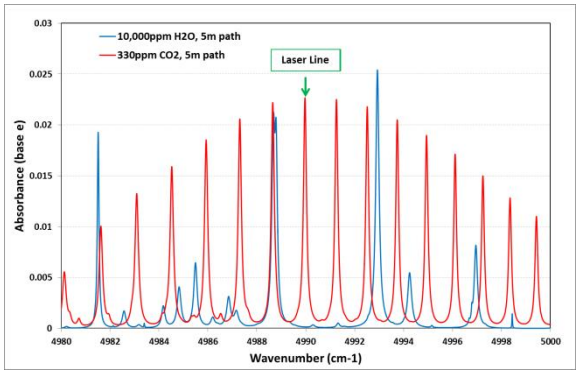
(a) IR Laser Control Module **(b) Transceiver**

Figure 13. Interiors of Modified RMLD components: (a) IR laser control module, and (b) transceiver.

Using PSI control board software, the modified detector (shown in Figure 13, hereafter referred to as the Remote CO_2 Leak Detector, RCLD), was tuned in laser operating temperature and current by monitoring the absorption of ambient CO_2. The laser was set to operate at 90 mA and 22.9 °C, which corresponds to an operating wavelength of 2.004 µm (or 4990 cm^{-1} in wavenumbers), overlapping a CO_2 absorption feature. This operating condition was selected in order to avoid the strong H_2O absorption features at 4989 and 4993 cm^{-1}. The modulation current depth was set at 17 mA corresponding to a wavelength sweep range of approximately 0.2 nm (or 0.5 cm^{-1}). Like the OPS, the RCLD was not calibrated to provide absolute measurement of CO_2, but comparison with LI-COR instrument measurements described below provides the information needed to complete such a calibration. They show that the reported ppm-m values should be multiplied by (1/0.55 = 1.82).



(a) Laser temperature as a function of laser current for operation at 2.004 µm (or 4990 cm^{-1}).

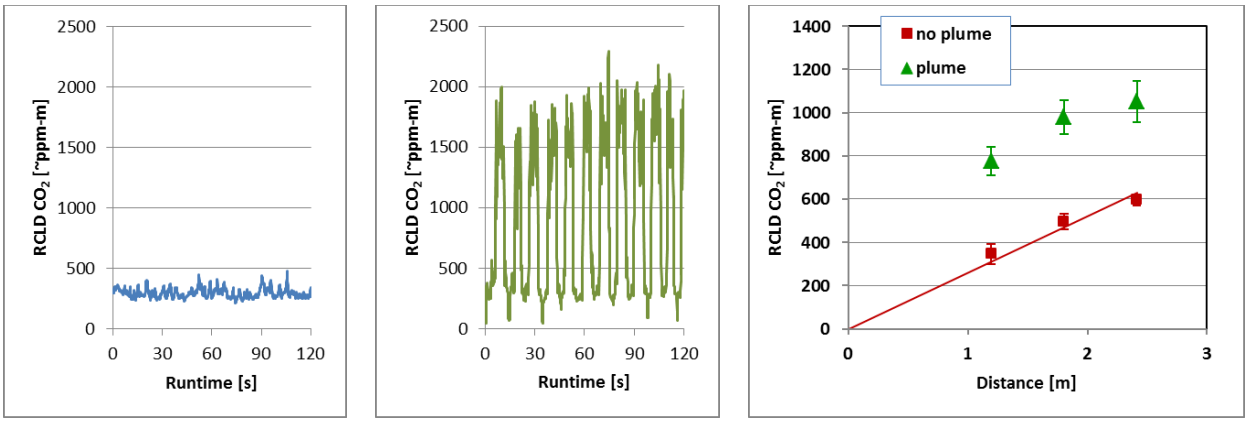


(b) CO_2 and Water absorbance spectra used to select the RCLD spectral line [7].

Figure 14. Tuning data for the RCLD with laser modulation about the 2.004 µm absorption in CO_2.

B. RCLD Lab Testing

On April 25, 2013, the RCLD was shipped to the ISGS for laboratory testing and field deployment. Throughout May 2013, laboratory tests were performed to characterize the RCLD responses. The RCLD response was evaluated by sweeping a jet of CO_2 (supplied from a high pressure bottle) perpendicular to the transceiver optical path. The charts in Figure 15 show typical responses of the RCLD to CO_2 jets set up within a fume hood.



(a) No flow **(b) CO_2 jet** **(c) vs. position**

Figure 15. Hand-held CO_2 detector response to CO_2 jets: (a) typical no flow case, (b) jet from a 0.101" dia. hole in a ½" PVC pipe, and (c) (peak) measurements of a 10 SLM jet from a 0.04" dia. hole vs. distance. The hand-held transceiver beam/field-of-view was waved periodically across the plume [(a), (b) from 5/17, (c) from 5/31].

C. RCLD Testing Timeline

Table 3 overviews laboratory and field testing of the RCLD conducted by ISGS personnel. A variety of laboratory tests of the RCLD were conducted at the ISGS, recording the data on a Windows PC or laptop. The responses to CO_2 jets, as well as the response as a function of distance to a fixed target were measured. Six field tests were conducted at IBDP, with an initial verification of the remote operation for the device on May 6, 2013, exposure to elevated CO_2 concentrations from a routine pipeline blowdown

procedure on July 11, 2013, and later comparisons to the OPS and LI-COR LI-8100A at IBDP during three site visits in October 2013.

Table 3. Summary of RCLD Laboratory and Field Testing by ISGS Personnel.

Date(s)	Tasks
3/19 – 3/22/2013	ISGS Staff visited Andover, MA to train with PSI engineers and worked on RMLD modification, calibration, and initial operational tests, and obtained software for data acquisition
4/25/2013	RCLD received by ISGS
5/6/2013	Field Test at IBDP: measured ppm-m vs. distance from various reflectors (10 meter range)
5/13/2013	Laboratory Test at ISGS: tested data acquisition system and software, recorded response to exhalation
5/17/2013	Laboratory Test at ISGS: verified responses to CO_2 jet in fumehood, measurements vs. distance (5 m)
5/28/2013	Laboratory Test at ISGS: took scans of CO_2 jet with selected flow rates from no flow up to 20 standard liters per minute (slm) of CO_2
5/31/2013	Laboratory Test at ISGS: took scans of 10 slm CO_2 jet with varied position to reflector
7/11/2013	Field Test at IBDP: measured exhaust plume from bleeder valve during routine pipeline blowdown
7/29/2013	Laboratory Test at ISGS: repeated 5/17/13 test, with data acquisition on laptop
10/8/2013	Field Test at IBDP: compared to OPS and LI-8100A, measurements vs. distance with various background targets up to 15 m
10/14/2013	Field Test at IBDP: compared to OPS and LI-8100A, measurements vs. distance with various background targets up to 25 m
10/16/2013	Field Test at IBDP: compared to OPS and LI-8100A, measurements vs. distance with various background targets up to 25 m
10/23/2013	Field Test at IBDP: compared to OPS and LI-8100A, measurements vs. distance with various background targets up to 25 m

D. RCLD Field Testing

An initial deployment of the RCLD was made on May 6, 2013 to test operation, and handling in the field, with additional deployments at IBDP in July to measure CO_2 from a pipeline blowdown procedure during injection startup operations, and in October to make comparisons with the OPS and LI-COR LI-8100A.

Measurements vs. Distance with Various Reflectors

The RCLD was field tested at the IBDP injection well site, collecting CO_2 measurements as a function of distance from various target surfaces. These surfaces include corrugated metal trailer exterior walls, container and vehicle doors, concrete barriers, and the reflective Stop sign used as the OPS target. Examples of these data are shown in Figure 16. The reflective stop sign (OPS target) provided a strong return signal (F1, not plotted) from ranges exceeding 20m, yielding a path-integrated CO_2 measurement that is linearly proportional to standoff distance. The other diffuse scattering surfaces, as expected, suffered relatively weak return signals at standoff distances exceeding about 5m. With these weak but detectable signals, the RCLD is programmed to reduce noise at the expense of calibration accuracy while maintaining ability to detect leaks as indicated by changes in relative, rather than absolute, CO_2 concentration. Thus, at ranges >5m, the RCLD response does not increase linearly with standoff distance. Return signals are too weak to elicit response at ranges beyond 10-15m depending on the diffuse target surface. This is consistent with design expectations.

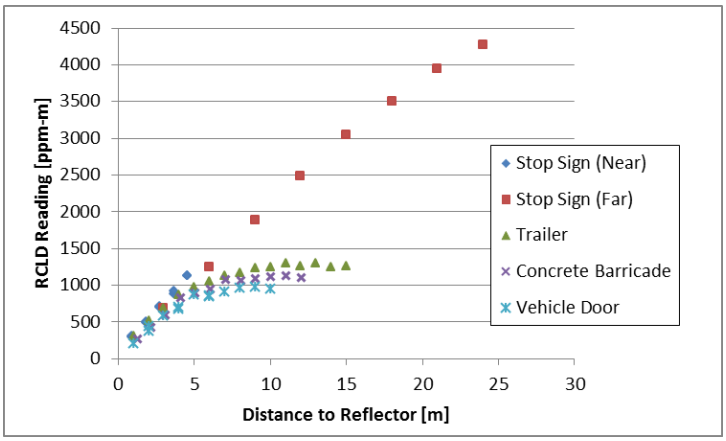


Figure 16. Measurements with various reflector targets (10/14/2013).

RCLD Measurements of CO_2 Plumes

On July 11, 2013, the RCLD was used at the IBDP site during maintenance work on the injection well. The hand-held detector was used to monitor the level of CO_2 emitted from the pipeline bleeder valve near the well head of CCS1, the IBDP CO_2 injection well. Aiming the RCLD at the valve from a distance of about 2.5m, baseline levels of 550-600 ppm-m were measured, while significant increases in concentration (2000-15000 ppm-m) were observed while the bleeder valve was open (see Figure 17).

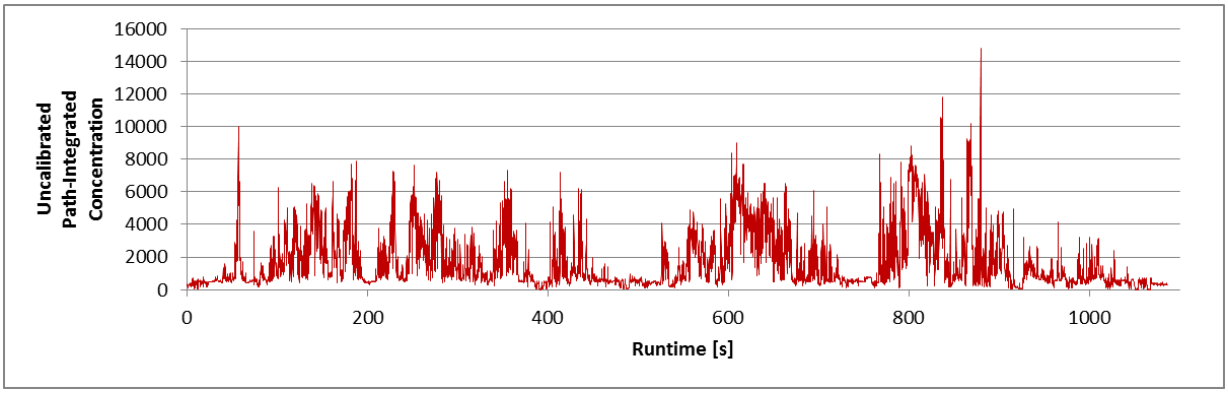


Figure 17. Measurements of CO_2 concentration using the RCLD to monitor CO_2 exhaust from the IBDP pipeline during routine maintenance operations.

Comparison to LI-COR LI-8100A (Atmospheric Mode) Measurements

During October 2013, CO_2 concentration measurements were made with both the RCLD and a LI-COR LI-8100A Soil CO_2 Flux System at the IBDP injection well. The LI-COR measurements and RCLD measurements are compared in Table 4. The LI-COR values shown are the average of three measurements taken along the OPS path, while the RCLD values are determined by the slope of the RCLD reading as the absorption path is increased (up to 15 feet or 4.6 m), using the OPS (stop sign) reflector as a target. The manufacturer stated measurement accuracy for the LI-8100A is 1.5% (or approximately 6.2 ppm for the results shown). The RCLD consistently reads approximately 55% of the value of the LI-8100A. This value provides an absolute calibration for the RCLD data.

Table 4. Comparison of LI-COR LI-8100A with PSI RCLD readings near IBDP injection well.

Date	Time (approx.)	LI-8100A [ppm]	RCLD [ppm]	RCLD/LI-8100A *100 [%]
10/08/2013	13:24	423		
10/08/2013	14:09		230	
10/08/2013	14:13		229	
10/08/2013	14:16		230	
10/08/2013	15:04		263	
10/08/2013	15:40	412		
10/08/2013	15:46		250	
10/14/2013	12:49		226	
10/14/2013	14:42		234	
10/16/2013	11:10	416	228	54.8%
10/16/2013	12:38	415	231	55.7%
10/23/2013	10:36	418	233	55.7%
10/23/2013	11:38		228	
10/23/2013	12:48	414		

VI. STATUS OF SENSORS

A. OPS

On November 8, the OPS was operational at the IBDP site. The current plan is to conduct a fully autonomous test of the system beginning November 15, and to retrieve the device from the field on December 16. Further deployment plans for the OPS are being discussed.

B. IGS

The majority of components for the IGS sensor head are currently at the ISGS in Champaign, Illinois. The malfunctioning laser components were returned by PSI to the manufacturer for analysis in June 2013. During the Summer 2013, PSI worked with staff at Heath Consultants to develop a power supply module, but this was not delivered to the ISGS with the other components, and no specifications were provided. Additional construction of this configuration and any future field deployment would be subject to future funding availability specifically for that purpose. The key issue is acquisition of sufficiently robust VSCELs that could be used to complete the construction and be field tested.

C. Remote Carbon-Dioxide Leak Detector (RCLD) Status

The RCLD was purchased by the ISGS. The device is operational, and will continue to be tested in the laboratory and field to support ongoing IBDP activities. Future testing will evaluate and quantify leak detection capabilities at 5-10m standoff range.

VII. CONCLUSIONS

This report summarizes the investigation of three prototype tunable diode laser absorption spectroscopy (TDLAS) sensors for carbon dioxide concentration measurements that have potential use in environmental monitoring programs at CCS sites. Two of the three sensor configurations were tested and deployed at the Illinois Basin – Decatur Project, achieving encouraging results about their suitability for leak detection.

The PSI Open-Path Sensor has withstood nearly 11 months of deployment at the IBDP site, monitoring carbon dioxide levels at the active injection wellhead. While some problems were encountered, the majority of OPS system components proved to be effective for use as an alarm-type system intended for personnel safety. The deployment was straightforward and required only standard installation needs (e.g., transceiver, reflector, and solar panel mounting). The transceiver housing, proved to be effective, keeping the electronics and optics free of moisture, while allowing easy access of components for analysis. Communications between the transceiver and the control computer were easily established with the radio modems properly configured, and data-logging with the supplied software was manageable, with a typical data rate of 91 kB/hour for 0.1 Hz sampling. The 19.3 AH Li-Ion battery used in the OPS has been maintained throughout deployment, and did not require replacement.

Two key problems were addressed during the OPS deployment: 1) The 30-W solar panel selected for charging the transceiver battery was undersized for some of the weather conditions (cloud cover) faced in the spring and fall at the installation location. This was remedied by replacing the panel with a 65-W option from the same manufacturer in October 2013, which to date has operated as desired even in poor weather; 2) Thermal stresses degraded optical alignment, likely due to the laser launch assembly mounted on the transceiver telescope. Thermal expansion of the telescope body apparently causes the laser to aim off center of the target reflector, resulting in diminished and occasionally lost return signal. This issue caused some loss of data during deployment at IBDP, and has reportedly been corrected by PSI in a methane OPS installed at natural gas leak detection test site. In addition, to make the OPS a more robust and accurate monitoring tool, calibrating the instrument signal (and recorded data) to relatively high accuracy measurements by other equipment should be part of standard procedures for future installations and operation.

The construction and deployment of the in-ground sensor could not be completed due to failure of the specialized infrared VCSEL components used in its design. The two VSCELs acquired by PSI were inoperable prior to delivery to the ISGS. Despite this problem, the hardware for the IGS sensor head was delivered to the ISGS with full assembly plans and instructions. Also, preliminary deployment plans for the sensor were discussed, including plans to acquire a power and communications tower from Heath Consultants, using similar batteries and radio modems to the OPS device. A replacement VCSEL device was not acquired given failure modes, lack of availability of an alternative supplier, and time constraints. The failure of the VCSELs during laboratory testing at PSI is a major concern, and it is perhaps necessary to consider some other options for the IGS laser component. Despite the higher laser cost, a redesign using the butterfly-style fiber-coupled laser may be a suitable technical option, considering the success achieved with applying this configuration in both the OPS and the RCLD prototypes.

The RCLD performed well during this field testing program. It was responsive to CO_2 plumes in both the laboratory setting, and in field demonstrations during the routine release of CO_2 from the pipeline bleeder valve near the well head. The device was demonstrated using surfaces of various objects as reflectors, at distances up to 25 m. The testing showed that the RCLD responds appropriately to CO_2 plumes in its optical path (demonstrated in concentration vs. time as the transceiver beam path is passed periodically over the plume), and that the device responds appropriately as the distance to the reflector is increased. Because the RCLD is intended as a leak detection tool and not a quantitative measurement tool, by design it computes concentration inaccurately when receiving weak return signals. This effect was observed when aiming the RCLD at diffuse backscatter targets at ranges exceeding 5m. Similar to the OPS,

calibration of the RCLD signal to other CO_2 measurement equipment should be included as part of its standard operating procedure.

The collaborative investigations of Physical Sciences Inc. and the Illinois State Geological Survey for this STTR program have illustrated the uses of TDLAS-based sensors. More specifically, two types of sensor configurations were tested successfully as CO_2 monitors at the IBDP site. With refinements, similar systems could be used at other CCS sites as part of an integrated environmental monitoring strategy.

VIII. REFERENCES

[1] Midwest Geological Sequestration Consortium (MGSC) website (www.sequestration.org) 2011. This material is based upon work supported by the U.S. Department of Energy under award number DE-FC26-05NT42588.

[2] M. Frish, et al., "Precision and Accuracy of Miniature Tunable Diode Laser Absorption Spectrometers." Physical Sciences Inc., Report SR-1446.

[3] *Personal communication*, March-April 2013, M. Frish, M. Laderer, Physical Sciences Inc., Andover, MA.

[4] *Personal communication*, April 15 2013, Steve Chancey of Heath Consultants, Houston, TX (via email conversation with M. Frish, Physical Sciences Inc., Andover, MA)

[5] *Personal communication*, August 16 2013, Christoph Gréus of VERTILAS GmbH, Garching/Munich, Germany (via email conversation with M. Frish, Physical Sciences Inc., Andover, MA)

[6] Heath Consultants Remote Methane Leak Detector (RMLD) User's Manual

[7] *Personal communication*, March 25 2013, R. Wainner, Physical Sciences Inc., Andover, MA.

Appendix A: Summary of Open-Path Sensor Measurements

A set of charts showing the CO_2 concentration measurements taken with the PSI Open-Path Sensor (OPS) over the course of the field deployment is given below.

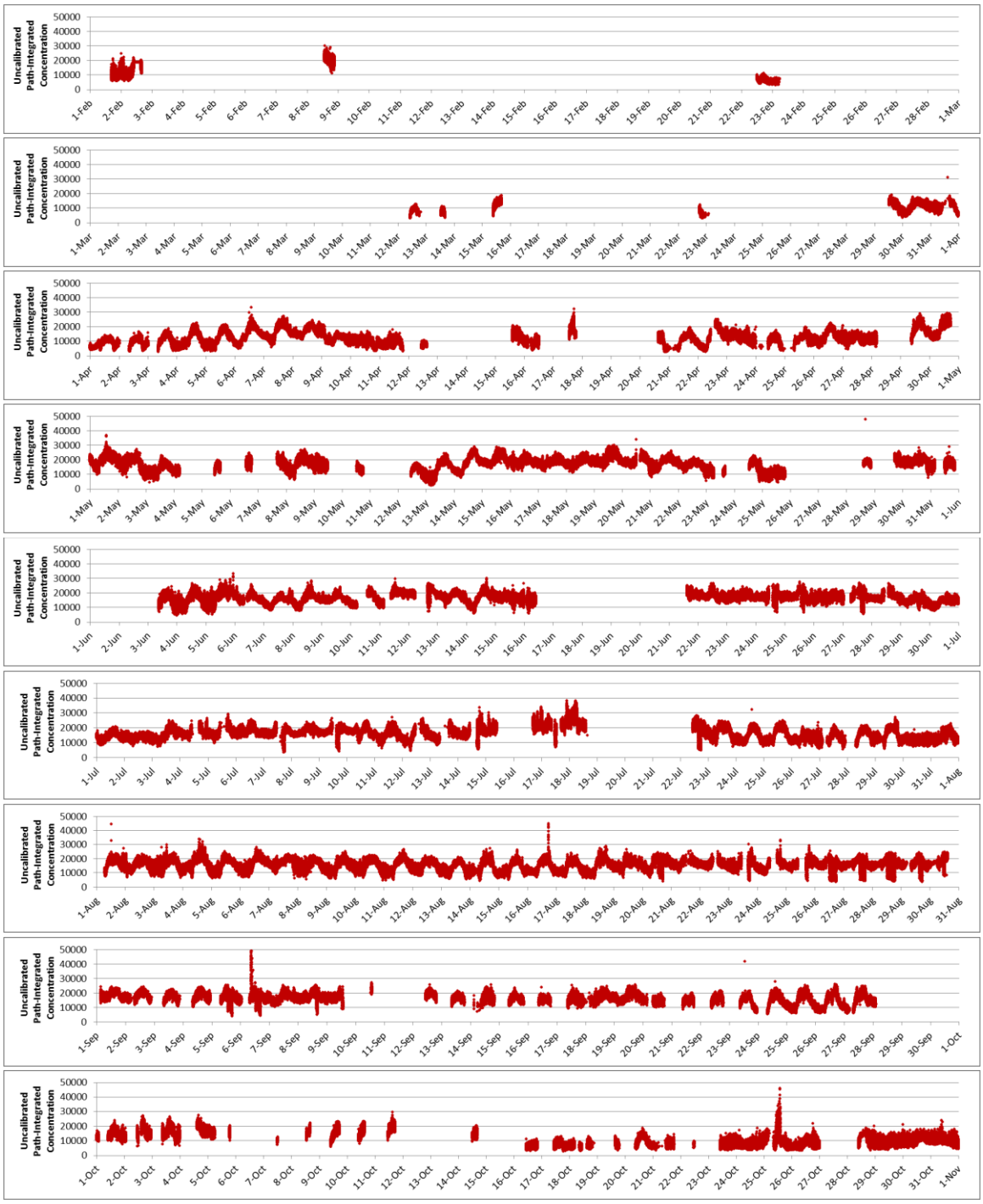


Figure 18. OPS CO_2 concentration measurements (filtered) February – October 2013. (NOTE: These values have not been corrected by other techniques for measuring atmospheric CO_2 concentrations. Atmospheric concentrations at this location are expected to be between about 400 and 600 ppm.)

Appendix B: Thermal Effects on Laser Alignmnet

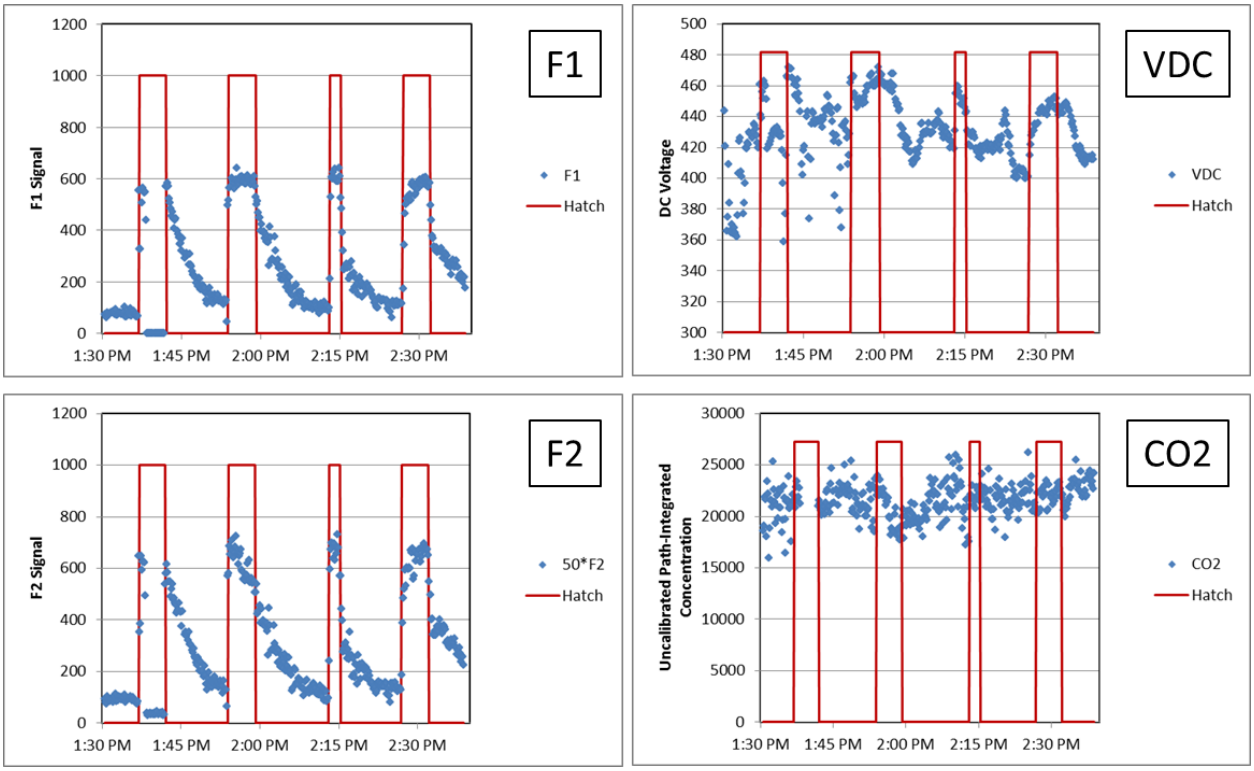


Figure 19. Influence of temperature on alignment. The temperature inside the enclosure is reduced when the hatch is open, during which the laser is aimed optimally at the reflective target, manifested as maximum F_1 signal. (During the first opening, the beam was replaced with the red laser for visual reference). After the hatch is closed, the enclosure heats up, and the laser aim is sub-optimum, resulting in a drop in both F_1 and F_2. Despite this drift, the sensor properly calculates the (uncalibrated) CO_2 concentration unless F_1<~50.

Appendix C: OPS Measurements during Routine Pipeline Blowdown

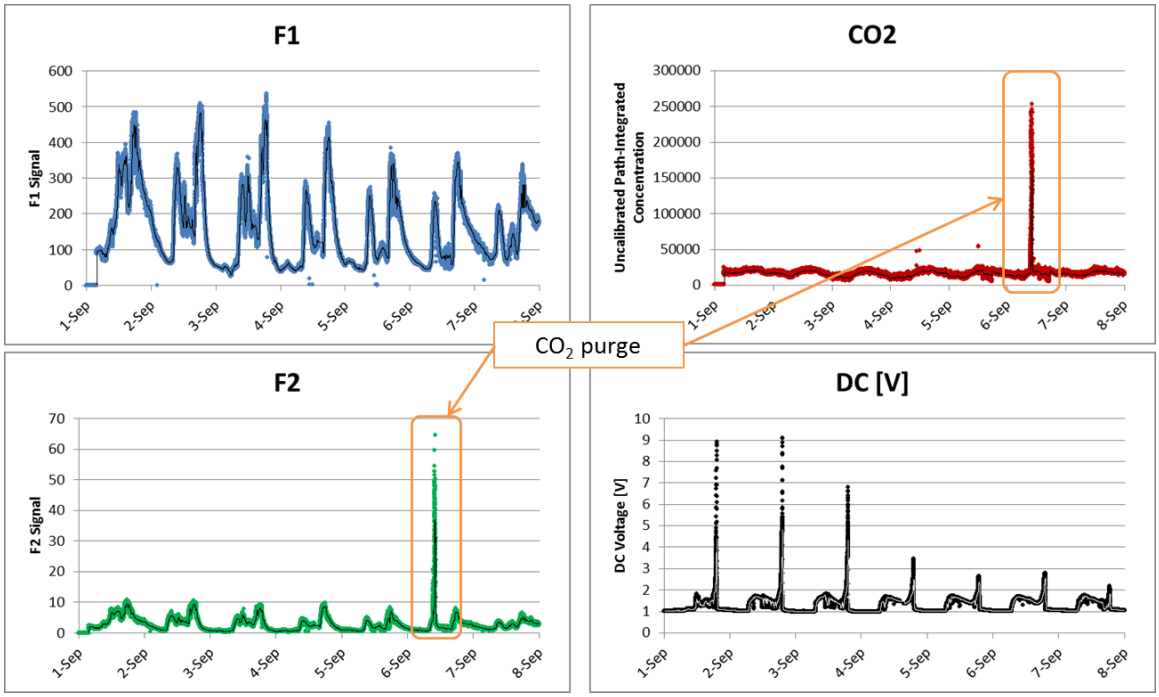


Figure 20. OPS internal signals and CO_2 measurement during IBDP pipeline blowdown (9/2013).

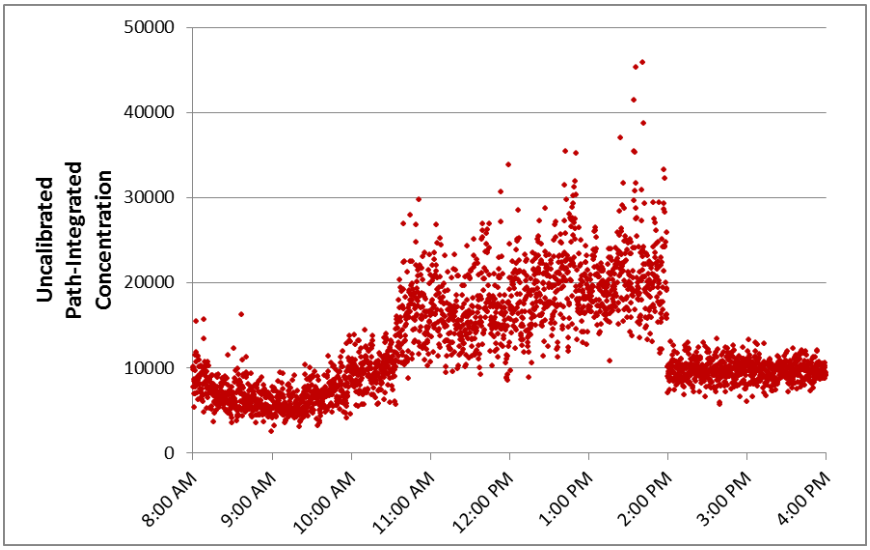


Figure 21. OPS CO_2 measurement during IBDP pipeline blowdown (10/25/2013, 10:30 AM to 2:00 PM).